Progress Report 3

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Project: Smart Door Anomaly Detection & Cybersecurity for Elderly Care
Course: CSIS 4495 – Applied Research Project, Section 3
Reporting Period: March 3 – March 9, 2026
Report Date: March 9, 2026

Work Date/Hours Logs

Date	Number of Hours	Description of work done
Mar 4, 2026	2.0	Reviewed fall detection API endpoints (/api/fall/detect, /api/fall/events, /api/fall/status, /api/fall/reset) and existing frontend codebase | Added FallEvent, FallEventsResponse, FallStatusResponse types and getFallEvents(), getFallStatus(), resetFallDetector() functions to frontend/lib/api.ts
Mar 6, 2026	1.5	Added "Falls Detected Today" stat card to frontend/app/page.tsx dashboard; updated grid from 4 to 5 columns
Mar 7, 2026	2.0	Created frontend/app/falls/page.tsx — fall history page with detector status badge, confidence bar visualization, event table, reset button
Mar 8, 2026	1.5	Added Falls nav item to frontend/components/Sidebar.tsx | Updated api/routes.py /api/stats endpoint to include falls.today count; updated StatsResponse interface in api.ts | Pushed code to main repo
Mar 9, 2026	1.0	Report making | Team meeting

Total Hours: 8.0

Summary Description of work done during this reporting period

This week, I integrated the fall detection system with the frontend dashboard. I reviewed the existing API endpoints and frontend codebase, then added TypeScript types and API client functions for fall detection. I enhanced the main dashboard with a "Falls Detected

Today" stat card and created a dedicated falls page with detector status, confidence visualization, event history table, and reset functionality.

Navigation was updated to include the falls section, and the stats API was modified to include fall counts. This completes the user interface for monitoring fall detection events in real-time.

The integration ensures caregivers can easily monitor fall incidents and manage the detection system through an intuitive web interface.

Repo Check in of Implementation completed

The files I have checked in the repo are as follows:

- `frontend/lib/api.ts` — Added fall detection types and API functions
- `frontend/app/page.tsx` — Added falls stat card to dashboard
- `frontend/app/falls/page.tsx` — New fall monitoring page
- `frontend/components/Sidebar.tsx` — Added falls navigation
- `api/routes.py` — Updated stats endpoint with fall counts

All changes committed via pull request with integration testing.

AI Use Section

AI Tool Name	**Version, Account Type**	**Specific feature for which the AI tool was used**	**Value Addition**
GitHub Copilot	Free Tier	TypeScript interface generation and React component code for fall detection UI	Customized for our fall event data structures and added proper error handling
ChatGPT	Pro Tier	UI design patterns for real-time monitoring dashboards	Adapted patterns for fall detection with appropriate visualizations and user interactions

Appendix: Prompt History

Prompt 1: "How to add TypeScript types for REST API responses in Next.js?"

- Used: Type definition patterns
- Modified: Created fall-specific interfaces for events and status responses

Prompt 2: "Best practices for displaying real-time data in React dashboards"

- Used: State management and update patterns
- Modified: Implemented for fall detection with confidence bars and status badges

Prompt 3: "Designing navigation for multi-feature web applications"

- Used: Sidebar navigation patterns
- Modified: Added falls section with appropriate icons and routing</content>